

Remapped Interfaces: Building Contextually Adaptive User Interfaces with Haptic Retargeting

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ABSTRACT

User interfaces in virtual reality are dynamic and contextual, making it challenging to provide accurate haptic feedback. This paper presents a remapped interface system that leverages haptic retargeting to reuse a limited set of controls for a wide array of interfaces. The system enables the creation of virtual user interfaces that can adapt to the context in which they are being used and provide accurate haptic feedback. Using the system, we developed four exemplar applications: an immersive painting app, an extensible music production tool, a simulated aircraft cockpit and a user-configurable interface design tool.

CCS CONCEPTS

• **Human-centered computing** → **Human computer interaction (HCI)**; **Virtual reality**; *Haptic devices*; • **Computing methodologies** → *Perception*.

KEYWORDS

haptics, <https://www.overleaf.com/project/637714d8a0c4d976810b6397>
control, redirection, interaction, perception, user interfaces, virtual reality

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1 INTRODUCTION

This paper presents a modular, remapped physical-virtual interface system and hardware prototype that supports the creation of rich remapped physical-virtual interfaces for Virtual Reality (VR). Remapped physical-virtual interfaces enable the creation and real-time reconfiguration of virtual interfaces with real haptic feedback. These interfaces can adapt to the context in which they are being used and retain real haptic feedback by mapping multiple virtual controls to a single, matching physical control.

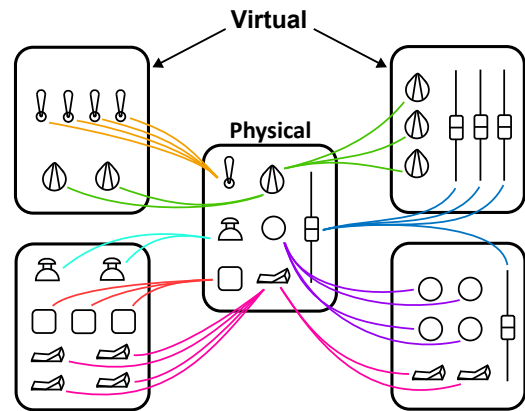


Figure 1: Using our system, the virtual controls on adaptive virtual interfaces are mapped to a limited set of physical controls.

When operating an *ideal* remapped physical-virtual interface, a user could naturally reach for and operate any control and receive realistic haptic feedback. They could, for example, grasp a slider and move it until they hit the upper or lower limit where the slider physically cannot move further. The layout of the virtual interface could change entirely, adapting to the context in which it is being used, and the user could again operate another slider in a different position and receive the same realistic haptic feedback. Figure 1 visualizes how our system maps multiple virtual interfaces onto a single physical interface. This functionality is achieved using haptic retargeting, creating the illusion that multiple physical controls exist.

The system presented in this paper is a step towards realising this type of interaction. We present an immersive experience that leverages the remapped interface system in multiple ways. The experience involves multiple unique interfaces that the user can select from. In addition, we present a system where users can reconfigure the interface in real-time to suit their individual preferences and needs.

2 RELATED WORK

A key benefit of user interfaces in virtual reality is their ability to adapt to the context in which they are being used and simulate hands-on interactions. While these interfaces are highly adaptive, the inability to directly touch them and receive realistic haptic feedback can break the user's feeling of presence and decrease interaction performance [3, 15].

Visuo-haptic illusions leverage the phenomenon of visual dominance within the human perceptual system [8] to change the physically perceived shape of virtual objects [2, 17]. Haptic retargeting [1] is a type of visuo-haptic illusion that can manipulate the user's perceived hand position during reaching actions, also known as hand redirection [16].

Haptic retargeting enables the reuse of passive haptics by mapping them to multiple virtual objects. In the body warping technique, angular and translational gains are applied to the user's hand as they move such that the user can touch one physical object while believing they are touching one of many multiple virtual objects [1].

Prior explorations combining passive haptics with virtual user interfaces have found they significantly improve user performance [10]. Other approaches have leveraged the user's non-dominant hand [9], or arm [5] to provide approximated passive haptic feedback, and this has been combined with haptic retargeting for even more flexibility [6].

Haptic retargeting has previously been applied to user interface interactions by Matthews et al. where a single button on a handheld controller was extended to nine virtual buttons [14]. Later, Feick et al. explored visuo-haptic illusions that can extend linear translation and presented a system to support multiple virtual sliders of different lengths using haptic retargeting and one motorized physical slider [7]. This paper extends upon these ideas to support the creation of rich, adaptive, remapped user interfaces. These interfaces provide both realistic haptic feedback while also being capable of adapting to the context within which they are being used.

3 REMAPPED INTERFACE SYSTEM

The remapped interface system enables virtual reality interfaces to adapt to the context in which it is being used while retaining the haptic feedback of a real physical control panel. Using haptic retargeting, a vast array of virtual interface layouts are remapped onto a physical proxy control panel with a selection of generalized, widely used input controls. Specifically, this system supports the remapping of sliders, push-buttons, knobs, switches, and joysticks. The physical controls communicate with the remapped interface system via wired or wireless communication. The physical controls can be arranged into a variety of layouts.

3.1 Physical and Virtual Interface Tools

The prototype physical interface used for this system is constructed of laser-cut MDF with 3D-printed brackets. The interface is modular, enabling limited reconfiguration of the available physical controls. Figure 2 shows the physical prototype interface with the control types: toggle switch, knob, thumbstick, push-button, slider, continuous knob, and rocker switch. Except for the slider, controls are mounted using generic mounts, allowing them to be swapped around.

This limited reconfigurability enables the installation of multiple instances of a physical control if they are extensively used on the virtual interface layouts. In addition, the amount of warping required to achieve the remapping can be reduced by tailoring the physical layout to the general proximity of the virtual controls. Within the system, the physical controls are represented by virtual



Figure 2: The physical prototype interface mounted to a desk with various physical controls. The Touch controller calibrates and aligns the panel within the tracking space.

replicas of the physical controls. These watch the state information provided by the physical interface and replicate it within the virtual environment. These are hidden from the user for the haptic retargeting illusion to function.

A developer can create virtual replicas of the available physical controls to construct the virtual interface. These controls can be reconfigured into almost unlimited layouts. However, the controls should ideally be as close as possible to the corresponding physical control and aligned in the same orientation as the physical control.

When a virtual control is 'selected' by the user, they are automatically linked such that the virtual control copies the physical control state. Stateful controls, like switches, knobs and sliders, can create state conflicts. State conflicts occur when the selected virtual control state differs from the corresponding physical control.

To resolve these conflicts, actuated controls are used that automatically move to match the state of the virtual control. We use a 10cm motorized slide potentiometer for sliders (PSM01-082A-103B2), though switches and knobs require customized solutions. Motorized rotary potentiometers were considered for the knobs; however, they move very slowly. The part we tested (RK16812MG099) takes approximately 12 seconds to move from limit to limit (12 s / 300°). Instead, we used motorized encoders (FIT0481) to complete the same movement in <1 second (150 RPM, 0.4 s / 360°).

Off-the-shelf options were unavailable for the switch, so we created a custom actuating switch. The custom actuated switch uses a servo that pushes a lever attached to a 3D-printed toggle or rocker. This switch, in turn, pushes a micro-toggle switch to enable the detection of the switch state.

3.2 Haptic Retargeting Implementation

Our haptic retargeting implementation is based upon On-The-Fly [4] by Cheng et al. to enable target changing using a dwell pointer-based target selection method. The 'Primitive' based Shape Aware haptic retargeting [13] ensures alignment between the user's hand and the physical controls.

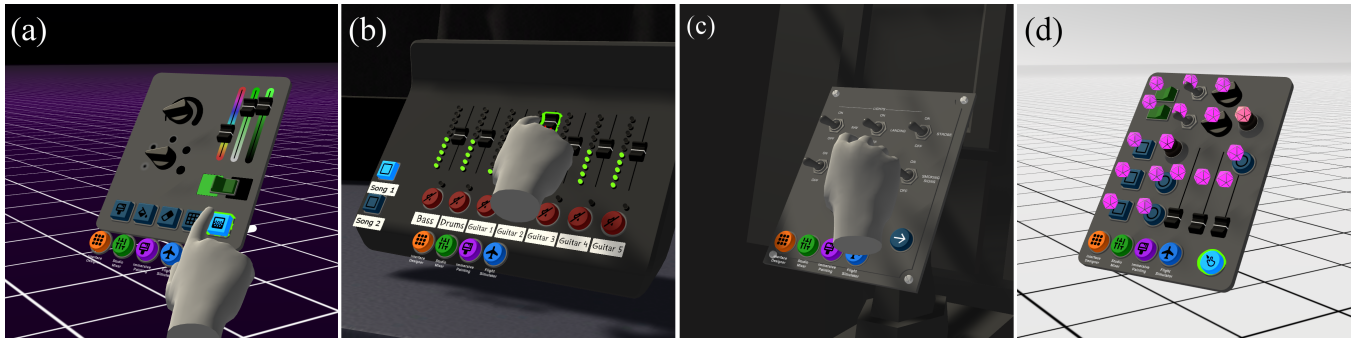


Figure 3: The virtual interfaces used in our four exemplar applications: (a) an interface for an immersive painting application, (b) an extensible Studio Mixer Interface with a song loaded with 7 channels, (c) a control panel in an simulated aircraft cockpit and (d) the interface editing tool in *editing* mode. In (d) the pink gems indicate the controls can be grabbed and rearranged by the user. The four buttons across the bottom of the interfaces allow users to swap between demos as they wish. experiences.

Haptic retargeting requires prior knowledge of the user’s intended target before their hand can be redirected. In this system, we use either head pointers or hand pointers. With the hand pointer, the user aims a pointer extending from their hand at the target. The head pointer uses a reticle that is directed wherever the user is looking [12]. Instead of a reset, the user is prevented from selecting controls that they are too close to. This forces them to retract their hand to select the next target but minimises the amount of reset required, similar to the adaptive reset techniques described by Matthews et al. [11]

4 EXEMPLAR INTERACTIVE APPLICATIONS

We present four exemplar applications, each using the same underlying physical interface remapped to different virtual interfaces, demonstrating the flexibility afforded by our system. The four examples are: 1) an immersive painting application, 2) an extensible music production system, 3) an immersive aircraft simulator, and 4) a user-configurable interface. Figure 3 shows the interfaces used in each exemplar. The user can also swap between experiences using four remapped buttons, also shown in Figure 3, demonstrating the adaptability of the interface.

4.1 Immersive Painting Application

The first exemplar is an immersive painting application inspired by experiences like Tilt Brush by Google¹. This demonstrates the applicability of remapped physical-virtual interfaces within a wide array of immersive experiences. In theory, any experience that involves these types of interfaces could incorporate these controls to provide a more immersive experience with realistic haptic feedback.

The user can configure their brush using the interface and then paint in the virtual world. For example, they can change the colour using Hue, Saturation and Value sliders. They can adjust the thickness and hardness of the line using knobs and swap between a solid colour and a gradient using buttons. Then the user can paint in 3D within the virtual world to create personalised artwork.

4.2 Extensible Music Production System

This exemplar demonstrates the extensible and modular nature of the remapped interface system. A collection of songs were selected, split into one channel for each instrument and the vocals. The user can select from a range of song by selecting one from the song list. Multiple channels are procedurally generated to match the number of channels in the selected song. The user can switch to another song, and the channels will automatically update to reflect the new song.

4.3 Immersive Aircraft Simulator

The remapped interface system can also simulate physical control panels in the real world, such as the controls on the flight deck of an aircraft. This exemplar demonstrates mapping a single control panel to multiple different fixed layouts. The user can swap the interface between an autopilot panel and a control panel with various switches to operate other utilities. This type of interface could be integrated with both commercial and hobby-grade simulators to improve the realism of the experience. Trainee pilots would be able to operate the physical controls of various aircraft within a generic physical simulator.

4.4 User-Configurable Interface Designer

Leveraging the flexibility of haptic retargeting, the immersed user can also enter an editing mode. In this mode, they can rearrange the controls into a new layout, similar to arranging toolbars in a desktop application. They can then return to the interaction mode and operate the interface with their new layout. To reconfigure the interface, the user can grab a control using a distance grab technique with a pointer. The controls remain attached to the surface of the virtual interface, and the user can move them within the interface’s boundary. Distance grabbing is used instead of grasping to prevent the user from inadvertently grabbing the physical controls. The distance-grabbing functionality was implemented using the Oculus Interaction System and Meta Quest hand tracking.

¹<https://www.tiltbrush.com/>

5 CONCLUSION

In this paper, we presented a novel physical-virtual interface system that facilitates the creation of a range of dynamically adjustable haptic experiences for VR. We presented the current state of our interactive system with several examples of how the tool can be used to create adaptable virtual user interfaces with realistic haptic feedback. The user can also reconfigure the interface in real time, highlighting the flexibility of the system for creating haptic user interfaces.

In the future, we would like to explore the use of this system in a comprehensive VR experience and gather data on acceptability and performance. We would also like to co-design potential applications within new domains to better understand the benefits and limitations, and potential uses of this novel technique.

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